UNIT 1: Trees

Pricing options; Europeans, barriers, Americans

Approximating diffusions: GBM and Local Volatility

Appendix: Lookback option

Trees as approximations of diffusions

- Approximate the risk-neutral dynamics of a diffusion process (such as GBM) using a discrete model with a finite number of branches from each state.
- The expectations (hence option prices) that we find in the tree approximate the expectations (hence option prices) under the diffusion process.
- For example, let us use a tree to approximate the diffusion

$$\mathrm{d}S_t = R_{grow} S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t$$

Use binomial tree to approximate diffusion dynamics

- Diffusion is

$$\mathrm{d}S_t = R_{grow} S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t$$

where W is $\mathbb{P}$-BM. So $X := \log S$ has dynamics

$$\mathrm{d}X_t = \nu \mathrm{d}t + \sigma \mathrm{d}W_t$$

where $\nu := R_{grow} - \sigma^2/2$.

- Approximate dynamics of $X := \log S$ using a binomial tree.

$$\mathbb{P}(X_{t+\Delta t} = X_t + \Delta x_u) = p$$

$$\mathbb{P}(X_{t+\Delta t} = X_t - \Delta x_d) = 1 - p$$

where Δx_u and Δx_d and p will be nonrandom parameters.

Match the means and variances

- Let $\Delta t := T/N$ where N is number of time steps.
- Now choose $p, \Delta x_u, \Delta x_d$ such that diffusion and tree agree on mean and variance of the increments. In the diffusion,

$$X_{t+\Delta t} - X_t = \int_t^{t+\Delta t} \nu \mathrm{d}s + \int_t^{t+\Delta t} \sigma \mathrm{d}W_s = \nu \Delta t + \sigma \Delta W.$$

So

$$\nu \Delta t = \mathbb{E}_t(X_{t+\Delta t} - X_t) = p\Delta x_u + (1-p)(-\Delta x_d)$$

$$\sigma^2 \Delta t = \mathrm{Var}_t(X_{t+\Delta t} - X_t) = p(\Delta x_u)^2 + (1-p)(-\Delta x_d)^2 - (\nu \Delta t)^2$$

Two equations and three unknowns: p, Δx_u, Δx_d. Can impose another condition (such as $\Delta x_u = \Delta x_d$ or $p = 1/2$) and solve. Then price the option in a tree with these parameters.

Not flexible enough

- However, often we want to specify Δx_u and Δx_d in advance.

  For example, we may want $\Delta x_u = \Delta x_d = \Delta x$, so that the *same* price levels are represented at every time point in the tree.

  Maybe, furthermore, we want to specify the size of Δx to optimize convergence or to choose *which* price levels are represented.

- This leaves only one free parameter p.

  Not enough to match both the mean and the variance.

Solution: trinomial trees.

Use trinomial tree to approximate diffusion dynamics

- Diffusion is

$$\mathrm{d}S_t = R_{grow} S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t$$

where W is $\mathbb{P}$-BM. So $X := \log S$ has dynamics

$$\mathrm{d}X_t = \nu \mathrm{d}t + \sigma \mathrm{d}W_t$$

where $\nu := R_{grow} - \sigma^2/2$.

- Approximate dynamics of $X := \log S$ using a trinomial tree.

$$\mathbb{P}(X_{t+\Delta t} = X_t + \Delta x) = p_u$$

$$\mathbb{P}(X_{t+\Delta t} = X_t) = p_m$$

$$\mathbb{P}(X_{t+\Delta t} = X_t - \Delta x) = p_d$$

Match the means and variances

- Let $\Delta t := T/N$ where N is number of time steps.

  Choose $\Delta x \approx \sigma\sqrt{3\Delta t}$ for accuracy reasons.

  Can modify these suggestions to make sure certain price levels or times are represented in the tree.

- Now choose p_u, p_m, p_d such that diffusion and tree agree on mean and variance of the increments. In the diffusion,

$$X_{t+\Delta t} - X_t = \int_t^{t+\Delta t} \nu \mathrm{d}s + \int_t^{t+\Delta t} \sigma \mathrm{d}W_s = \nu\Delta t + \sigma\Delta W.$$

  So

$$\nu\Delta t = \mathbb{E}_t(X_{t+\Delta t} - X_t) = p_u\Delta x + p_m 0 + p_d(-\Delta x)$$

$$\sigma^2\Delta t = \mathrm{Var}_t(X_{t+\Delta t} - X_t) = p_u(\Delta x)^2 + p_m(0)^2 + p_d(-\Delta x)^2 - (\nu\Delta t)^2$$

$$p_u + p_m + p_d = 1$$

Solve for probabilities

Solve system of three equations in three unknowns.

$$p_u = \frac{1}{2}\left[\frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2} + \frac{\nu\Delta t}{\Delta x}\right]$$

$$p_m = 1 - \frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2}$$

$$p_d = \frac{1}{2}\left[\frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2} - \frac{\nu\Delta t}{\Delta x}\right]$$

Intuition: For fixed Δt and Δx,

- The bigger the σ, the more probability mass in the wings, the less in the middle.
- The bigger the ν, the more mass in the up-branch, the less in the down-branch.

Option pricing

To find option prices, induct backwards in tree with time points

$$t_n = n\Delta t \qquad n = 0, \ldots, N,$$

and log-stock-price points

$$x_j = \log S_0 + j\Delta x \qquad j = -N, \ldots, N.$$

Option price in tree for node at time t_n and log-price x_j is

$$C_n^j = e^{-r\Delta t}[p_u C_{n+1}^{j+1} + p_m C_{n+1}^{j} + p_d C_{n+1}^{j-1}].$$

With large enough N, the option price in the tree $\approx$ the option price for diffusion, because in the $N \to \infty$ (or equivalently $\Delta t \to 0$) limit, the tree's option price $\to$ option price under diffusion dynamics.

For Europeans, the proof is by a form of the CLT.

Can't replicate general payoffs if 3 states 2 assets

- Recall that in the 3-state model, general options cannot be replicated using stock and bank acct. There is not a unique arbitrage-free price for the option. So how can we talk about finding "the" price of the option?
- Answer: The trinomial tree is not the model of the market.
  The model is GBM in continuous time. In that model, replication using $\{B, S\}$ is possible. So unique price exists.
  The tree is a *computational device* that we use in order to approximate the risk-neutral expectations (hence the prices) of the continuous-time model.

Option Pricing: Knock-outs and Americans

Pricing of knock-outs

$$C_n^j = e^{-r\Delta t}[p_u C_{n+1}^{j+1} + p_m C_{n+1}^{j} + p_d C_{n+1}^{j-1}]\mathbf{1}_{\text{no knock-out at time } t_n}$$

Pricing of American put

$$C_n^j = \max((K - S_{t_n}^j)^+,\ e^{-r\Delta t}[p_u C_{n+1}^{j+1} + p_m C_{n+1}^{j} + p_d C_{n+1}^{j-1}])$$

Formulas are similar to binomial tree.

Local volatility models

We want a model that is consistent with the observed non-constant implied vol skew. One approach: *local volatility* models specify the instantaneous volatility σ to be a function of (S_t, t).

- In the continuous-time setting,

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma(S_t, t)S_t\mathrm{d}W_t$$

  where W is $\mathbb{P}$-BM.

- In the tree setting, we let σ depend on the node (S_t, t).

Note that the local volatility σ is not the same thing as σ_{imp}.

Analytic computation of option prices is difficult in the diffusion setting, even for European calls/puts (exception: $\sigma(S_t, t) = \sigma(t)$). But by approximating the diffusion in a tree, the computations are easy.

Option pricing in the tree setting

Suppose we are given the local volatility function σ.

- Let $\Delta t = T/N$ (unless this fails to place important dates in tree).
- To choose Δx, let σ_{avg} be some "representative" or "average" σ in the tree, and let σ_{max} be an upper bound on the σ in the tree. Two guidelines: to make local discretization error small, we want

$$\Delta x \approx \sigma_{avg}\sqrt{3\Delta t}$$

but for stability reasons, we want

$$\Delta x \geq \sigma_{max}\sqrt{\Delta t}$$

So we can let $\Delta x = \max(\sigma_{avg}\sqrt{3\Delta t}, \sigma_{max}\sqrt{\Delta t})$

Option pricing in the tree setting

- Then at each node, use the $\sigma(S_t, t)$ prevailing at that particular node to generate the ν and the probabilities for the branches out of that node. Same formulas as L1:

$$p_{u,d} = \frac{1}{2}\left[\frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2} \pm \frac{\nu\Delta t}{\Delta x}\right], \quad p_m = 1 - \frac{\sigma^2\Delta t + \nu^2(\Delta t)^2}{(\Delta x)^2}$$

- Price options – including path-dependent and American-style options – as we did for GBM.

  The only change is that σ (hence ν, p_u, p_m, p_d) vary across nodes.

What if σ is not given

Then *calibrate* it to the prices of listed options.

The general idea of calibration:

- Observe prices of liquidly traded assets (e.g. listed options)
- Choose the model's parameters (e.g. the σ function) in such a way that the model generates theoretical prices that match closely the observed prices.

Then one can apply that model, with the calibrated parameters, to

- Compute hedges and risk sensitivities
- Price and hedge illiquid options
- Price and hedge complex contracts (e.g. structured products and exotic options)

Calibration of local volatility $\sigma(t)$

If σ is a non-random function of t and

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma(t)S_t\mathrm{d}W_t$$

then $\mathrm{d}\log S_t = (r - \frac{1}{2}\sigma^2(t))\mathrm{d}t + \sigma(t)\mathrm{d}W_t$ so

$$\log S_T = \log S_0 + \Big(r - \frac{\bar{\sigma}_T^2}{2}\Big)T + \int_o^T \sigma(t)\mathrm{d}W_t$$

$$\sim \mathrm{Normal}\Big(\log S_0 + (r - \frac{\bar{\sigma}_T^2}{2})T,\ \bar{\sigma}_T^2 T\Big) \text{ where } \boxed{\bar{\sigma}_T := \sqrt{\frac{1}{T}\int_0^T \sigma^2(t)\mathrm{d}t}}$$

(intuition: a nonrandomly weighted sum of indep normals is normal)

So time-0 call prices $C(K,T) = C^{BS}(\bar{\sigma}_T)$, thus $\sigma_{imp}(K,T) = \bar{\sigma}_T$.

Calibration: If given $C(K,T)$ at various expiries T, then obtain σ_{imp} and use boxed equation to (not uniquely) find $\sigma(t)$.

Calibration of local volatility $\sigma(S,t)$ in trinom tree

It's harder when σ depends on (S,t).

Here's the rough idea. Assume $r=0$ and a tree with S-levels s_j with equal spacing $\Delta S = s_{j+1} - s_j$. Let $\Delta K := \Delta S$.

Given: $C_0(K,\tau)$, the time-0 price of a strike-K expiry-τ call, for all (K,τ). Find at each node (s_j,t_n): Local volatility σ which generates risk-neutral probabilities consistent with call prices (and stock prices).

- The probability of reaching node (s_j,t_n) equals $(\Delta K)\mathrm{Fly}_0(s_j,t_n)$ where $\mathrm{Fly}_0(K,\tau)$ is the time-0 price of a *butterfly*:

$$\mathrm{Fly}_0(K,\tau) := \frac{C_0(K-\Delta K,\tau) - 2C_0(K,\tau) + C_0(K+\Delta K,\tau)}{(\Delta K)^2}$$

Calibration of local volatility in a trinomial tree

- The probability of reaching node (s_j, t_n) *and then going up* equals $(\Delta t/\Delta K)\mathrm{Cal}_0(s_j, t_n)$, where $\mathrm{Cal}_0(K, \tau)$ is the time-0 price of a *calendar spread*:

$$\mathrm{Cal}_0(K, \tau) := \frac{C_0(K, \tau + \Delta t) - C_0(K, \tau)}{\Delta t}$$

by a replication argument.

Hence the conditional up-probability from node (s_j, t_n) is the ratio

$$p_u = \frac{\mathbb{P}(\text{reach } (s_j, t_n) \text{ then up})}{\mathbb{P}(\text{reach } (s_j, t_n))} = \frac{(\Delta t/\Delta K)\mathrm{Cal}_0}{(\Delta K)\mathrm{Fly}_0}$$

Calibration of local volatility in a trinomial tree

On the other hand, the up-probability is also obtained by solving

$$\begin{aligned} p_u(\Delta K) + p_m(0) + p_d(-\Delta K) &= 0 \\ p_u(\Delta K)^2 + p_m(0) + p_d(-\Delta K)^2 &= \sigma^2 s_j^2(\Delta t) \end{aligned} \qquad \Rightarrow p_u = \frac{\sigma^2 s_j^2 \Delta t}{2(\Delta K)^2}$$

where $\sigma,\ p_u, p_m, p_d$ all depend on (s_j, t_n). Therefore

$$\frac{\sigma^2 s_j^2 \Delta t}{2(\Delta K)^2} = \frac{(\Delta t/\Delta K)\mathrm{Cal}_0}{(\Delta K)\mathrm{Fly}_0}$$

Conclusion: at node (s_j, t_n), the local volatility calibrated at time 0 is

$$\sigma(s_j, t_n) = \sqrt{\frac{2}{s_j^2} \times \frac{\mathrm{Cal}_0(s_j, t_n)}{\mathrm{Fly}_0(s_j, t_n)}}$$

Conceptually, the local volatility $\sigma(s, t)$ can be inferred from prices of options with strikes near s and expiries near t.

Calibration of local volatility in a trinomial tree

Comments:

- We have derived a discrete version of the "Dupire equation."
  We'll return to this formula in the PDE context.
- If you try to implement this formula directly, the calibrated σ will be sensitive to noise in the option price observations.
  In practice some regularization/smoothing procedure is advisable.
- Calibration is much easier in the case where σ depends only on t instead of jointly on (S, t).